

Automatic Detection of Parkinson's Disease Using Gait Signals and Time-Series Feature Engineering

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Subheadline möglich. Gegebenenfalls löschen.

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Motivation & Background

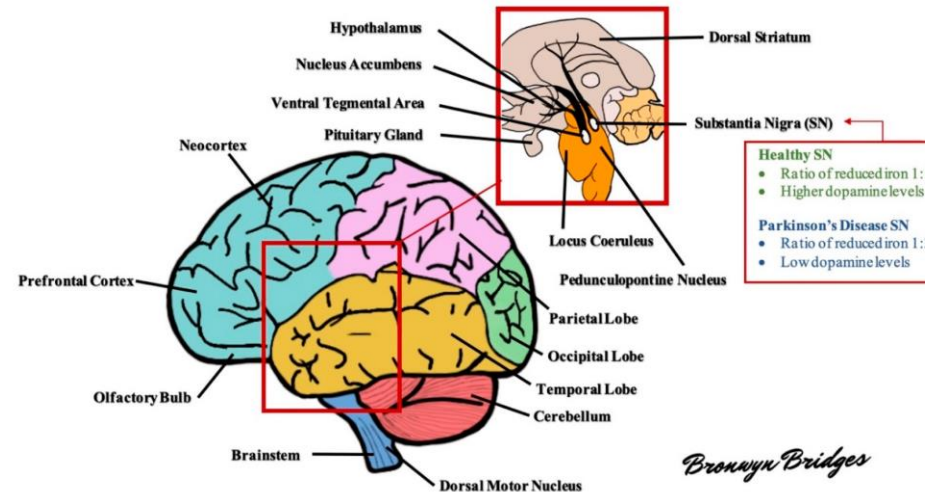


Image source: Bronwyn Bridges et al. (2020), *Molecules*, MDPI, <https://doi.org/10.3390/molecules25194382>

- ✓ Parkinson's Disease (PD) causes characteristic gait abnormalities, including altered rhythm, reduced stride length, and gait asymmetry.
- ✓ Vertical Ground Reaction Force (vGRF) signals captured during walking provide measurable indicators of these motor impairments.
- ✓ Time-series feature extraction (TSFresh) enables automated extraction of statistical and temporal descriptors from gait signals.
- ✓ Machine learning models can leverage these features to distinguish Parkinson's patients from healthy controls.
- ✓ This study evaluates classical machine-learning baselines for PD detection using gait signals from the PhysioNet GaitPDB dataset.

Dataset Description & Demographics

Group	Subjects	Mean Age ($\pm$ SD)	Male	Female
Control	72	63.7 $\pm$ 8.7	40	32
PD	93	66.3 $\pm$ 9.5	58	35

Figure: Participant demographics distribution by group, age, and gender.

- ✓ Total of **165 subjects** analyzed (72 Controls, 93 PD patients).
- ✓ **Mean age:** PD group is slightly older (66.3 $\pm$ 9.5 years) than Controls (63.7 $\pm$ 8.7 years).
- ✓ **Gender balance:** Both groups have comparable male, female ratios, ensuring unbiased comparisons.
- ✓ **Relevance:** These demographic similarities support reliable baseline feature and gait analysis.

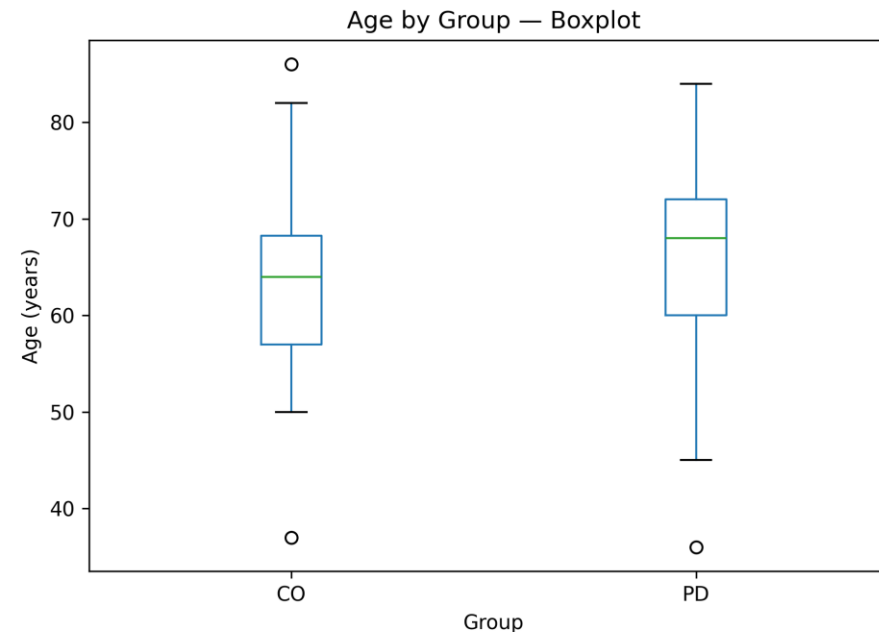


Figure: Age distribution by group

- ✓ **Dataset:** 165 subjects (≈ 93 PD + 72 Controls)
- ✓ **Signal type:** Vertical Ground Reaction Force (VGRF) from both feet
- ✓ **Sampling rate:** 100 Hz (10 ms intervals)
- ✓ **Observation:** PD group shows slightly higher median age but similar variability
- ✓ **Relevance:** Ensures demographic comparability between groups before model training

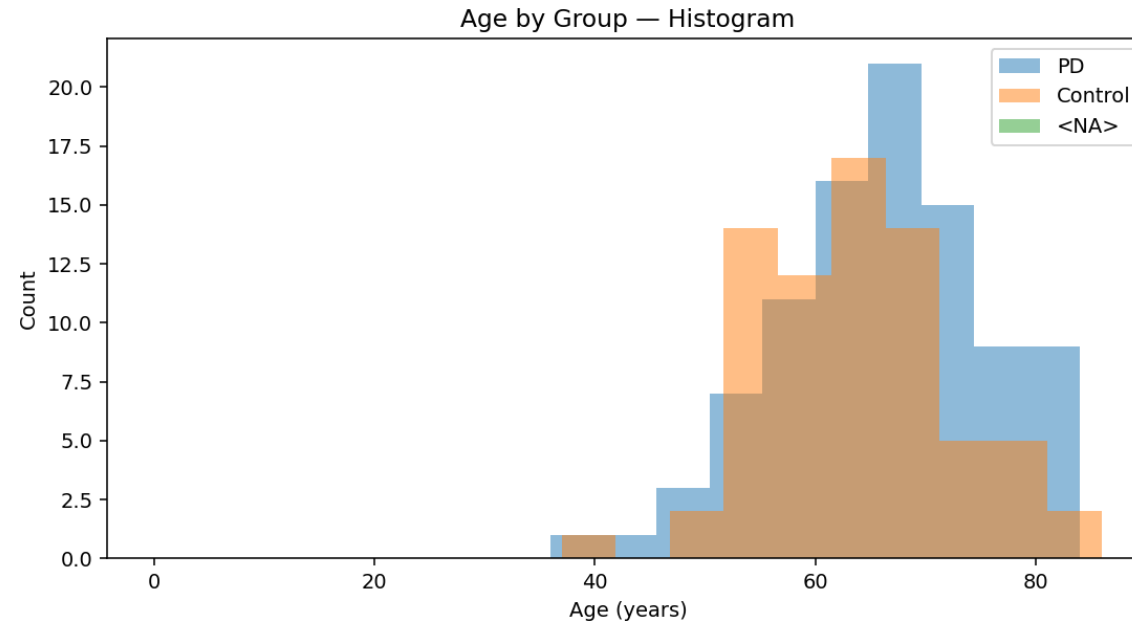


Figure: Age distribution histogram

- ✓ Visualizes age distribution across PD and Control subjects.
- ✓ PD participants show a higher frequency in the 65–80 year range.
- ✓ Both groups have comparable spread, supporting balanced dataset demographics.

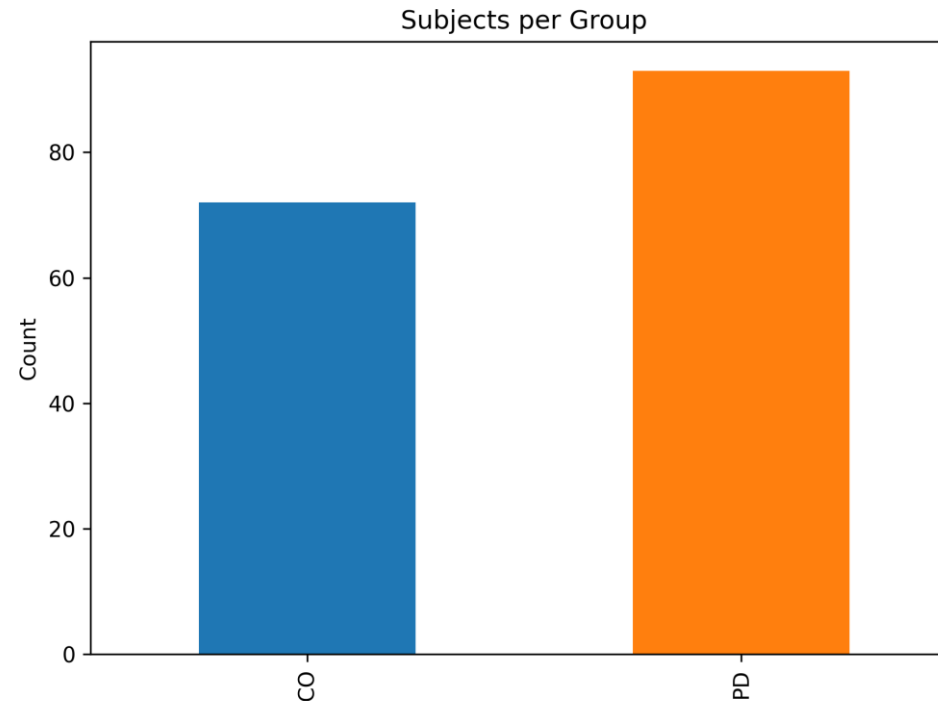


Figure: Subject distribution across PD and Control groups

- ✓ Visualizes participant count per group.
- ✓ **~72 Control** and **93 PD** subjects included in the final analysis.
- ✓ Slightly higher PD count, ensuring adequate representation for both classes.
- ✓ Balanced group sizes support fair model training and comparison.

Signal Examples (Control vs Parkinson's Patient)

Example VGRF Signals – Control Subject

Zoomed Vertical Ground Reaction Force (VGRF) – Healthy Control

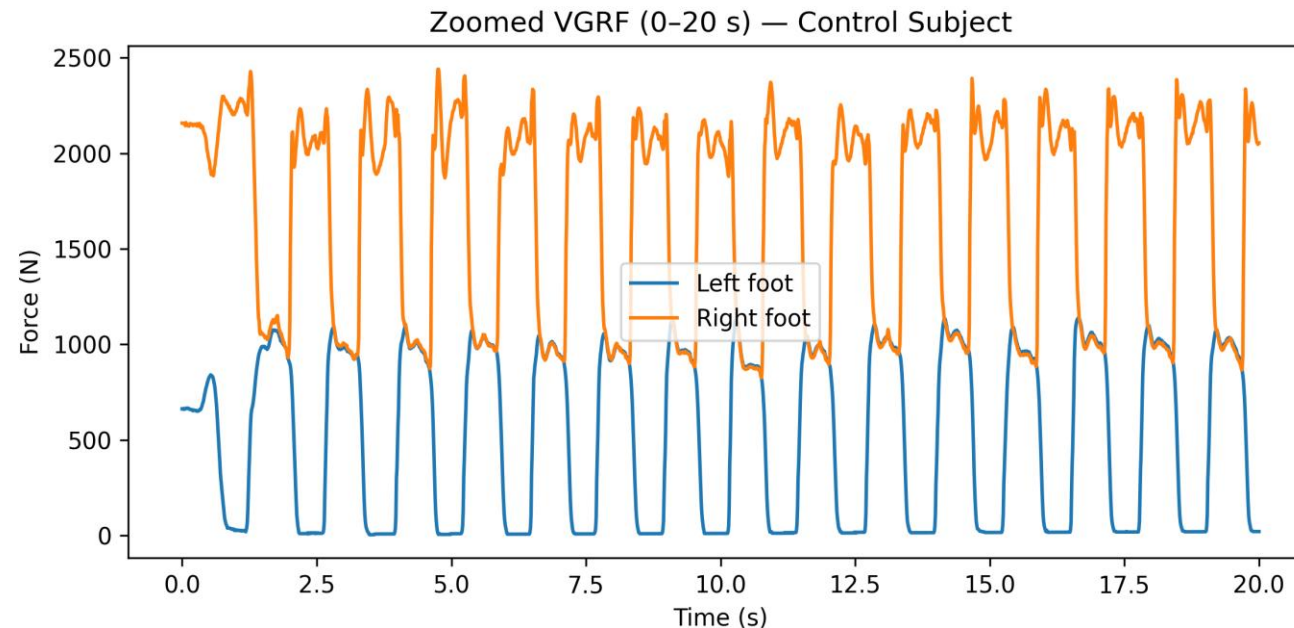


Figure: Control subject VGRF pattern (0–20 s segment)

- ✓ Displays periodic Vertical Ground Reaction Force (VGRF) signals for a healthy control subject.
- ✓ Left and right foot curves alternate rhythmically, showing regular gait cycles and balance.
- ✓ Both feet reach similar peak forces, indicating symmetry and stability.
- ✓ Small amplitude variations reflect normal step-to-step dynamics.
- ✓ Serves as baseline for comparison with Parkinson's patient signals.

Example VGRF Signals – PD Subject

Zoomed Vertical Ground Reaction Force (VGRF) – Parkinson's Patient

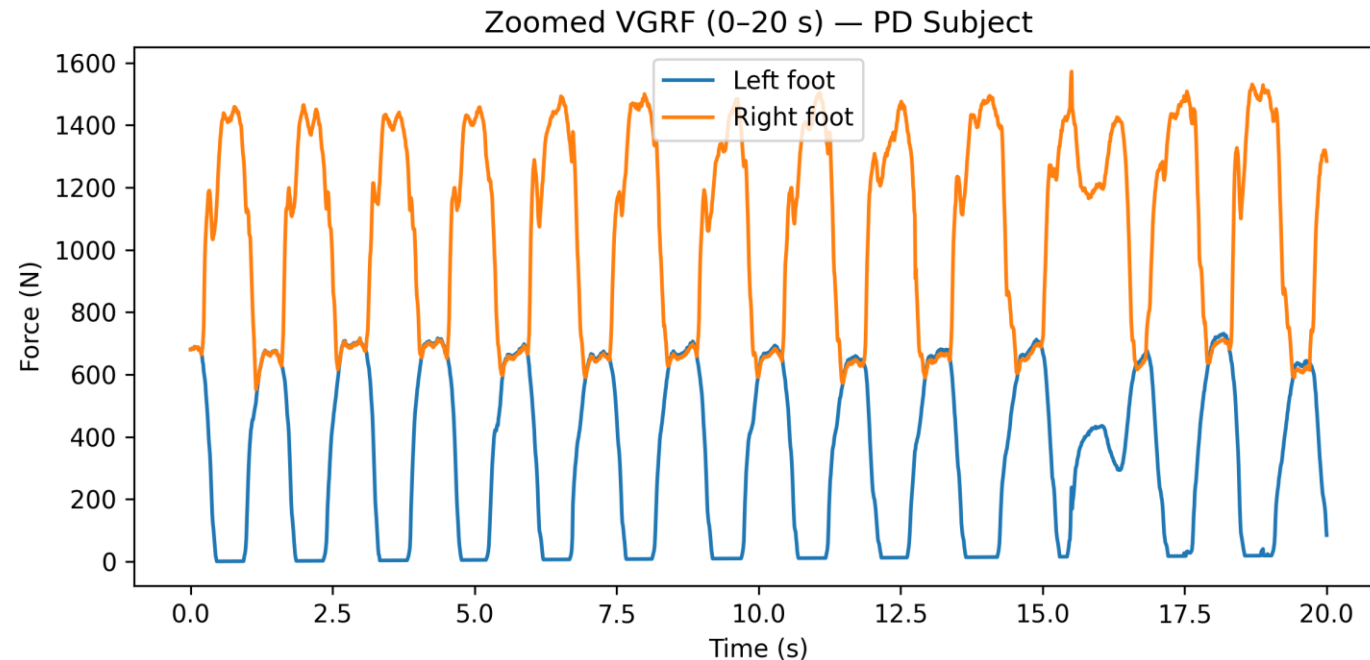


Figure: PD subject VGRF pattern (0–20 s segment)

- ✓ Shows VGRF for a Parkinson's patient during gait (same 0–20 s window).
- ✓ Left and right foot signals exhibit irregular amplitudes and timing differences.
- ✓ Gait cycles lack rhythm and symmetry indicating instability and asymmetry.
- ✓ Reduced and inconsistent peak forces suggest motor impairment and weak foot-ground interaction.
- ✓ Demonstrates the contrast with control pattern, capturing key PD-related gait abnormalities

Feature Extraction Approach

Feature Extraction Approach

Feature Extraction with TSFresh



Figure: TSFresh-based feature extraction workflow

Image source: Bronwyn Bridges et al. (2020), Molecules, MDPI, <https://doi.org/10.3390/molecules25194382>

- ✓ Raw VGRF time series are transformed into **statistical features** using **TSFresh**.
- ✓ Features summarize time-domain properties:
→ *mean, standard deviation, skewness, kurtosis, energy, RMS, zero-crossings, etc..*
- ✓ Extraction performed using a **30 ms window** with **15 ms overlap**.
- ✓ Each feature represents a short segment of the gait signal.
- ✓ Result: ~56 features per subject per foot after aggregation.

Feature Extraction Approach

Feature Combination Strategy

Feature Combination Strategy - Left & Right Foot Concatenation

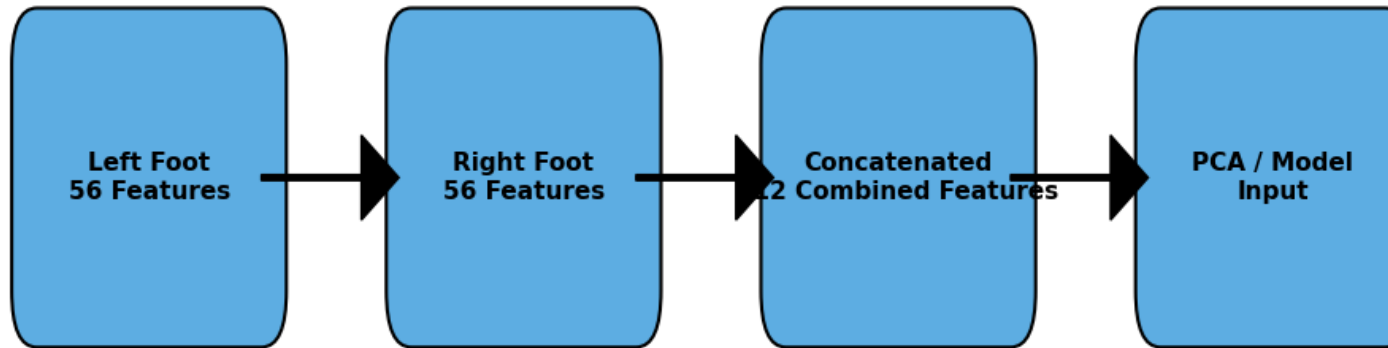


Figure: Feature Combination Strategy

- ✓ Each subject has two independent feature sets:
 - Left foot features (56 columns)
 - Right foot features (56 columns)
- ✓ To analyse inter-foot coordination:
 - Concatenated horizontally → 112 combined features (**LR vector**).
 - Two approaches tested:
 - ☐ **Concatenation:** merge L + R feature vectors.
 - ☐ **Averaging:** compute mean of corresponding left/right features.
- ✓ Combined feature set improves PD detection by capturing gait asymmetry.

Feature Analysis (Variance and Correlation)

Feature Analysis (Variance and Correlation)

Feature–UPDRS Correlation Analysis

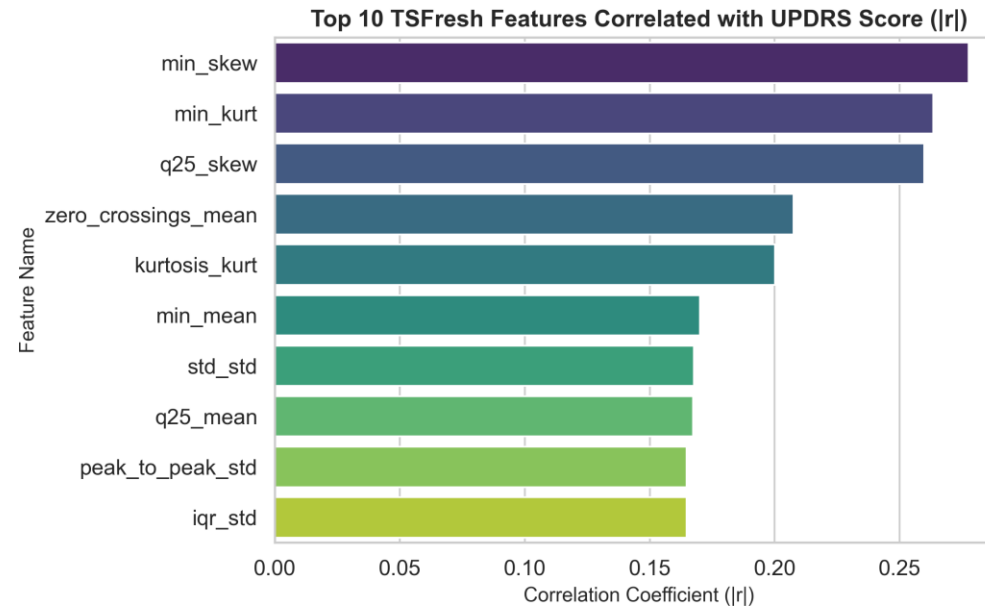


Figure: Feature–UPDRS Correlation Analysis

- ✓ Correlation computed between each extracted TSFresh feature and UPDRS score.
- ✓ UPDRS (Unified Parkinson's Disease Rating Scale) measures Parkinson's severity (higher = worse).
- ✓ Top features with higher $|r|$ values show stronger relationships with disease progression.
- ✓ Visualization methods:
 - *Heatmap* of feature–UPDRS correlations.
 - *Bar plot* of top 10 absolute correlations ($|r|$).
- ✓ Helps identify gait-related parameters (e.g., variability, symmetry) most sensitive to PD severity.

Feature Analysis (Variance and Correlation)-1

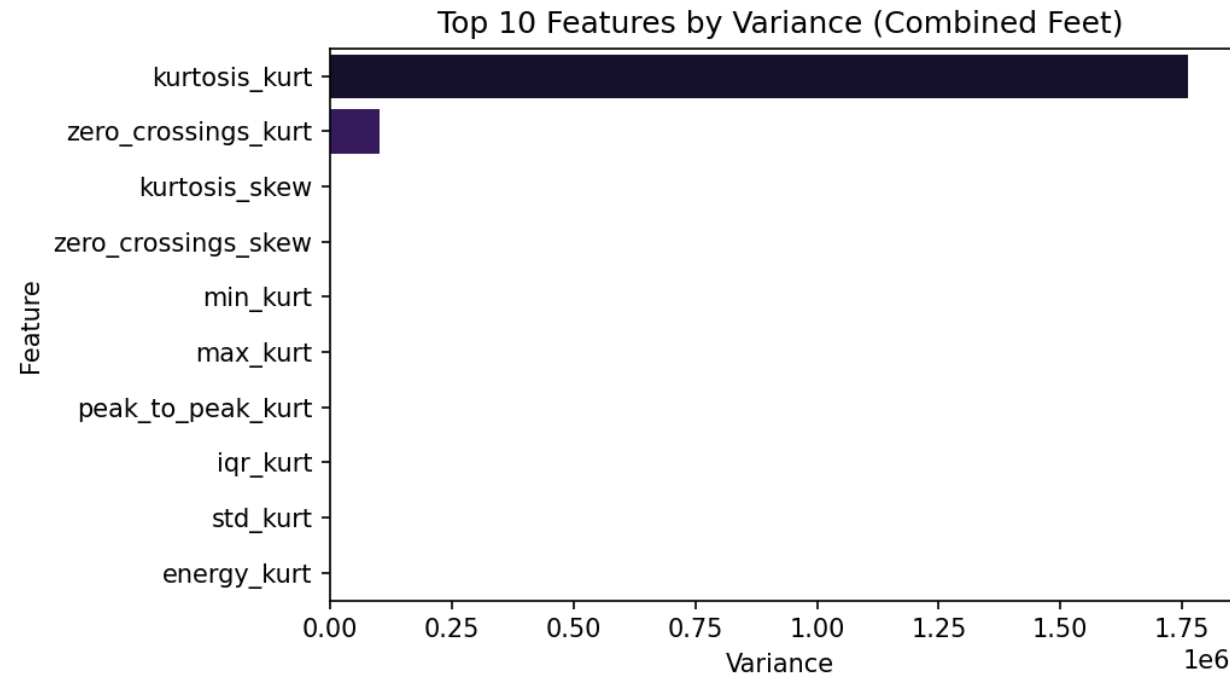


Figure: Top 10 high-variance features (combined feet)

- ✓ Displays the most variable TSFresh features from combined left–right foot data.
- ✓ *Kurtosis_kurt* shows the largest variance, suggesting strong sensitivity to gait irregularities.
- ✓ Variance ranking helps identify the most discriminative features for PD vs. Control.
- ✓ These top features are later used for dimensionality reduction via PCA.

Feature Analysis (Variance and Correlation)-2

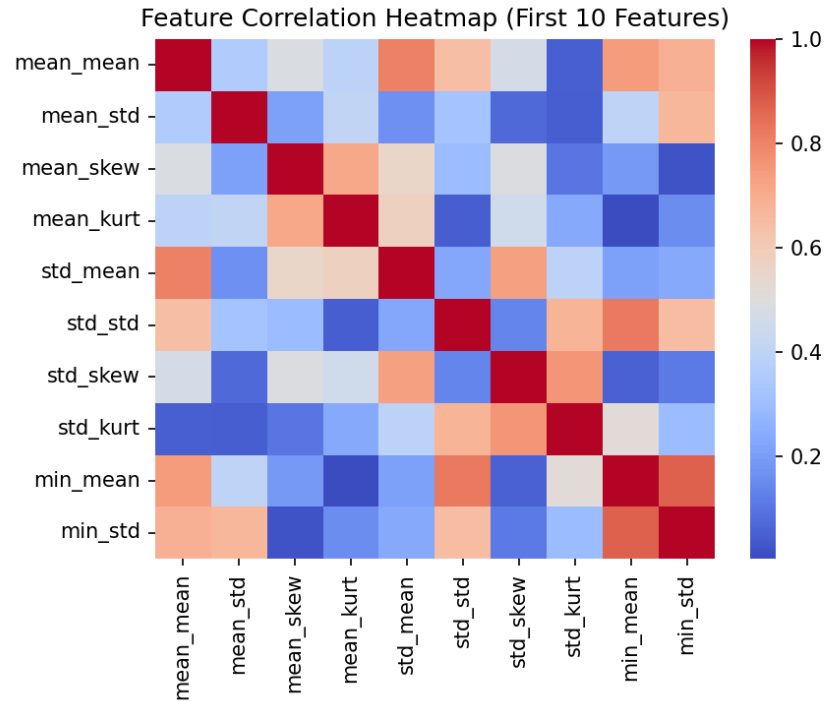


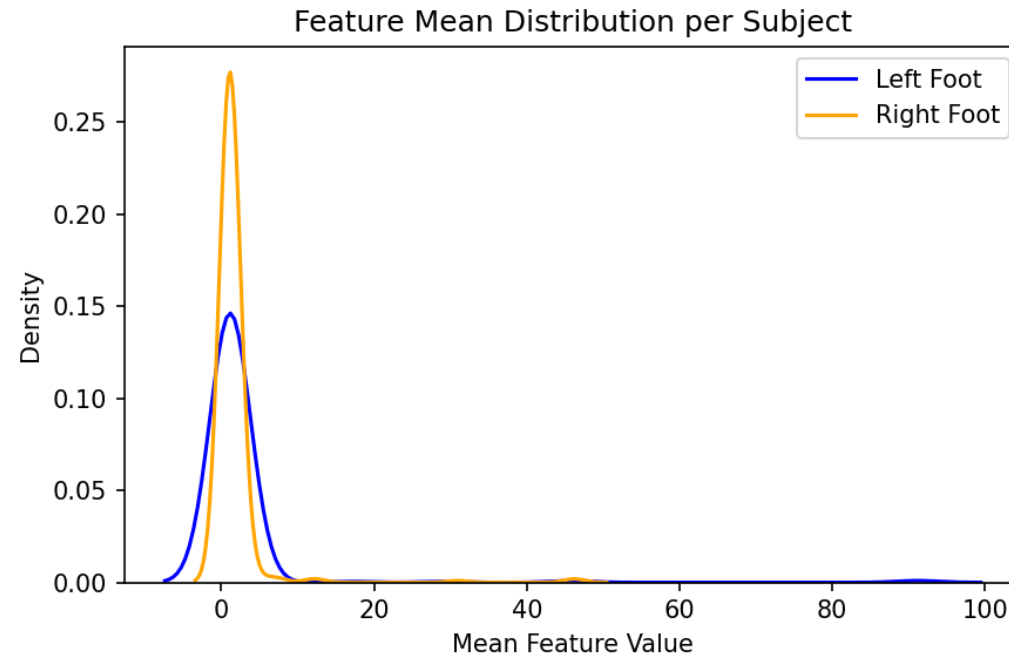
Figure: Correlation among top 10 features

- ✓ Displays correlation between the first 10 extracted TSFresh features.
- ✓ Moderate correlations observed between mean and standard deviation metrics.
- ✓ High feature redundancy is visible, justifying the use of PCA for dimensionality reduction.
- ✓ Ensures that strongly correlated features are compactly represented for efficient analysis.

Dimensionality Reduction (PCA Analysis)

Dimensionality Reduction (PCA Analysis)

Left vs Right Foot Feature Distribution



Feature mean distribution across left and right feet

- ✓ Compares mean feature distributions between left and right foot signals.
- ✓ Both feet show a similar overall distribution, confirming consistent gait capture.
- ✓ Slight right-foot dominance may indicate asymmetric force behavior in PD patients.
- ✓ Confirms that extracted features are balanced across sides before PCA analysis.

Dimensionality Reduction (PCA Analysis)

PCA Visualization – Right Foot Features

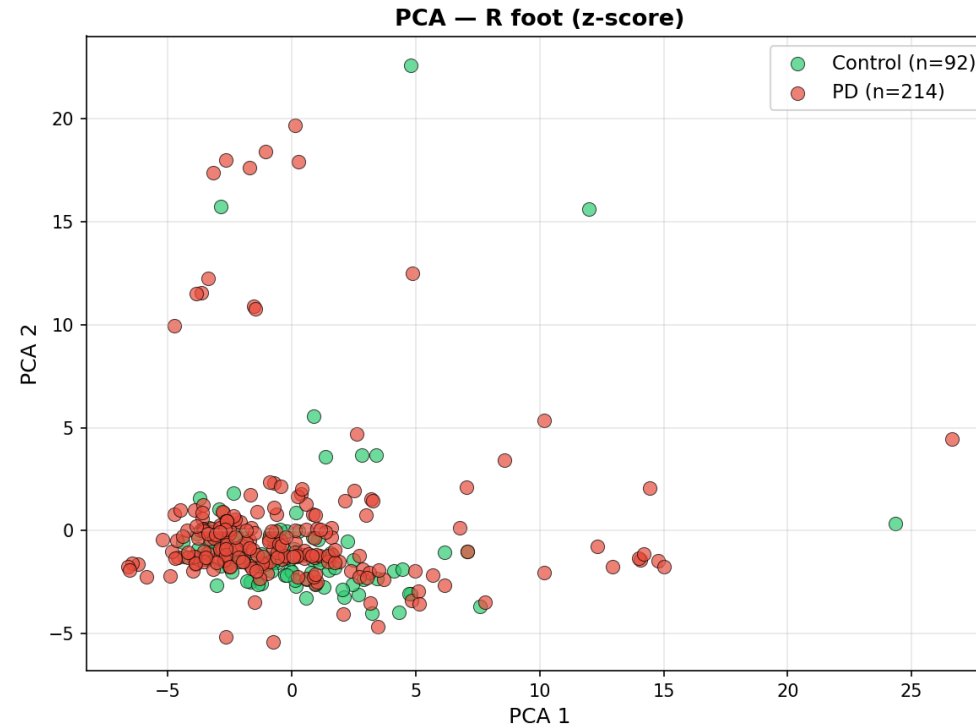


Figure: PCA projection for right-foot features

- ✓ PCA reduced 56-dimensional right-foot features to 2 principal components.
- ✓ Partial class separation is observed between PD (red) and Control (green) subjects.
- ✓ The right-foot dynamics reveal stronger gait irregularities in PD participants.
- ✓ Highlights that right-foot features may be more sensitive indicators of motor asymmetry.

Dimensionality Reduction (PCA Analysis)

PCA Visualization – Left Foot Features

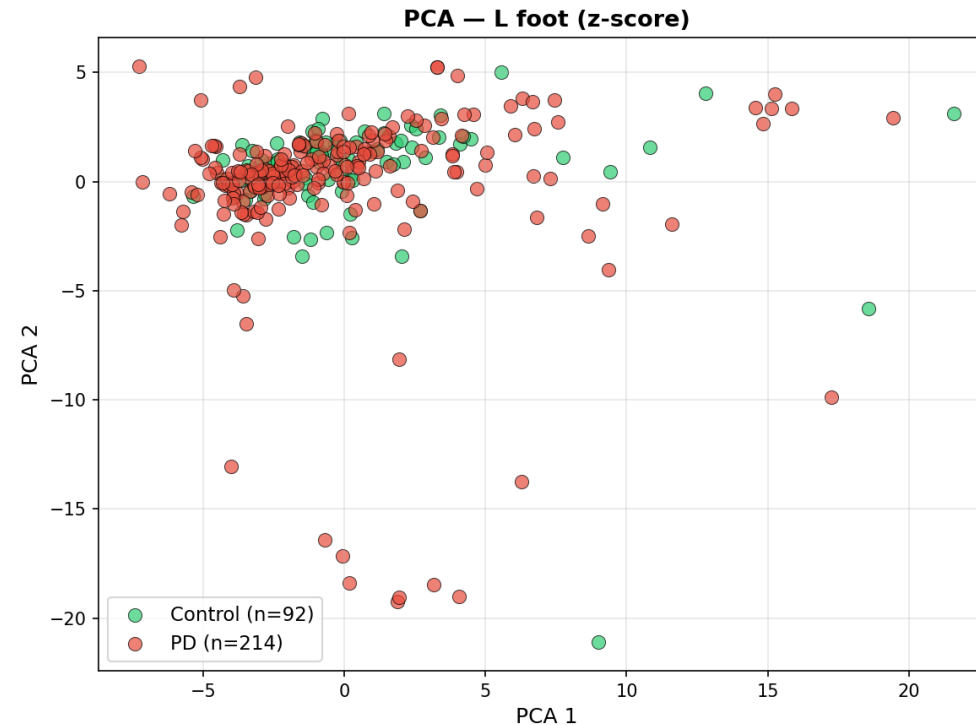


Figure: PCA projection for left-foot features

- ✓ PCA reduced 56 left-foot features to 2 principal components for visualization.
- ✓ Control (green) and PD (red) subjects show overlapping clusters with mild separation.
- ✓ Left-foot data appears slightly more compact, indicating consistent stride rhythm.
- ✓ Suggests that PD-related asymmetry is less pronounced on the left side compared to the right.

Dimensionality Reduction (PCA Analysis)

PCA Visualization – Combined Left & Right Features

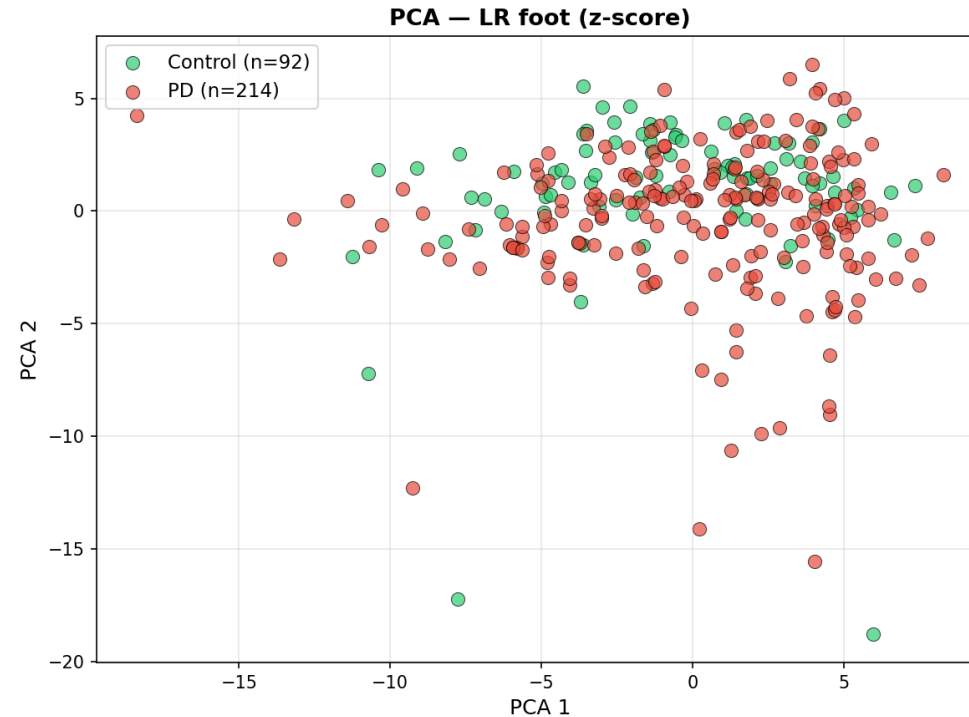
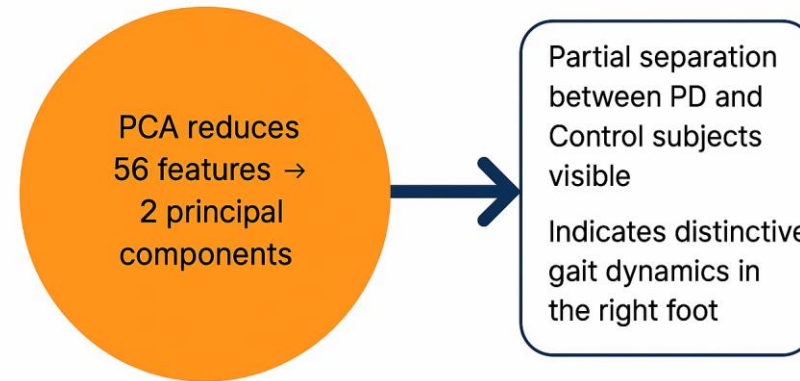


Figure: PCA projection for combined left & right foot features

- ✓ Combines left and right foot features to analyze overall gait symmetry.
- ✓ PCA reveals partially separable clusters between PD and Control participants.
- ✓ Integration of both feet provides richer movement representation and balance cues.
- ✓ This combined feature space provides the input representation used for subsequent machine learning classification.

Feature Analysis Results

Summary of Baseline Results



- ✓ **165 subjects analyzed** (93 PD patients, 72 controls).
- ✓ **56 TSFresh features extracted** from gait signals using **30 ms windows with 15 ms steps**.
- ✓ **Z-score normalization applied** before dimensionality reduction.
- ✓ **PCA reduced 56 features → 2 principal components** for visualization.
- ✓ **Partial separation observed** between PD and control groups.
- ✓ **Right-foot features showed stronger differentiation**, suggesting gait asymmetry.
- ✓ These findings **provide a reliable TSFresh-based feature representation for machine learning classification**.

TSFresh-Based Machine Learning Baselines for Parkinson's Disease Detection

After completing data exploration and feature analysis, we apply classical machine-learning models to classify Parkinson's disease using gait-derived TSFresh features. These models provide a strong baseline for evaluating the discriminative power of the extracted features and for understanding how well gait dynamics can distinguish Parkinson's patients from healthy controls.

Motivation for Baseline Models

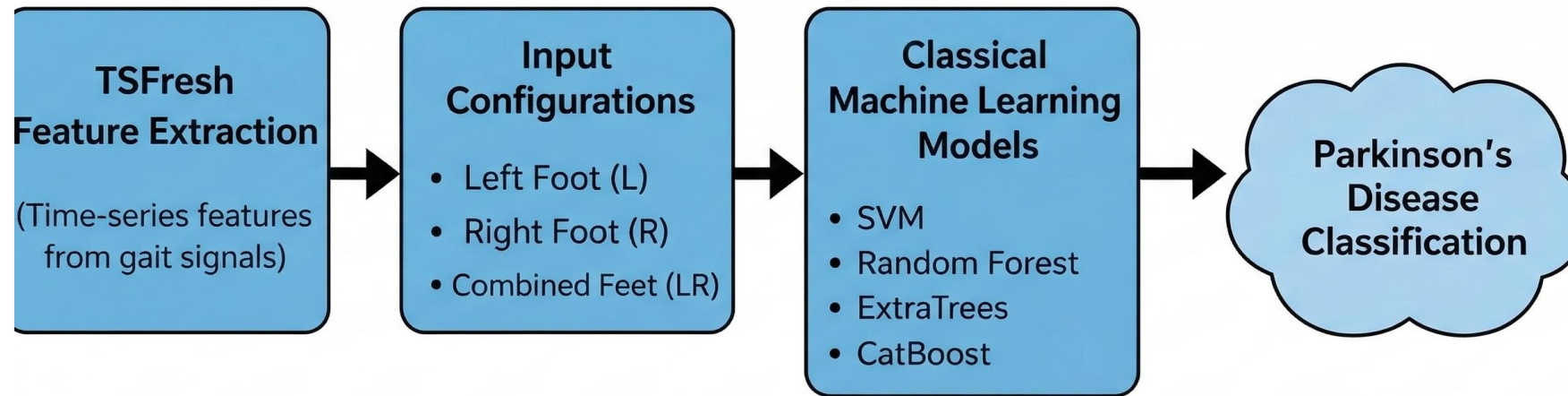


Figure: Baseline modeling workflow using TSFresh features for left, right, and combined gait inputs.

- ✓ **Strong baselines are essential** for evaluating machine learning performance.
- ✓ The objective is to **classify Parkinson's disease from gait signals** using feature-based models.
- ✓ **TSFresh features** provide a robust representation of gait dynamics.
- ✓ Baseline models establish a **quantitative reference for model comparison**.
- ✓ Models are evaluated **separately** on:
 - Left foot (L)
 - Right foot (R)
 - Combined feet (LR)

Models Used & Justification

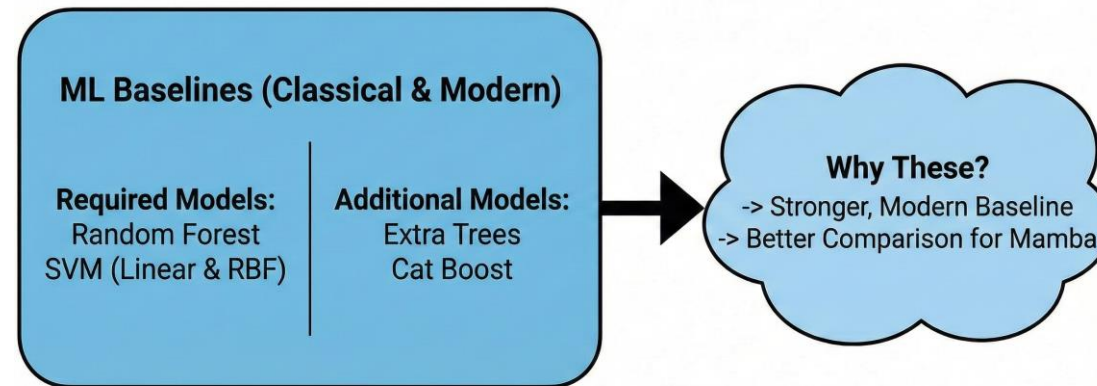


Figure: Overview of selected baseline models used for Parkinson's disease classification.

Required Models

- ✓ **Random Forest** — strong ensemble baseline for tabular data
- ✓ **SVM (Linear)** — linear decision boundary baseline
- ✓ **SVM (RBF)** — nonlinear kernel baseline

Why ExtraTrees and CatBoost?

- ✓ TSFresh produces high-dimensional, tabular features

Tree ensembles handle:

- Feature redundancy
- Nonlinear interactions
- Noise in gait signals

- ✓ ExtraTrees reduces variance compared to Random Forest and is robust for noisy biomedical data.
- ✓ CatBoost is a modern gradient-boosting algorithm widely used for tabular biomedical data.

Additional models selected:

- ✓ **ExtraTrees**
- ✓ **CatBoost**

Training & Validation Strategy

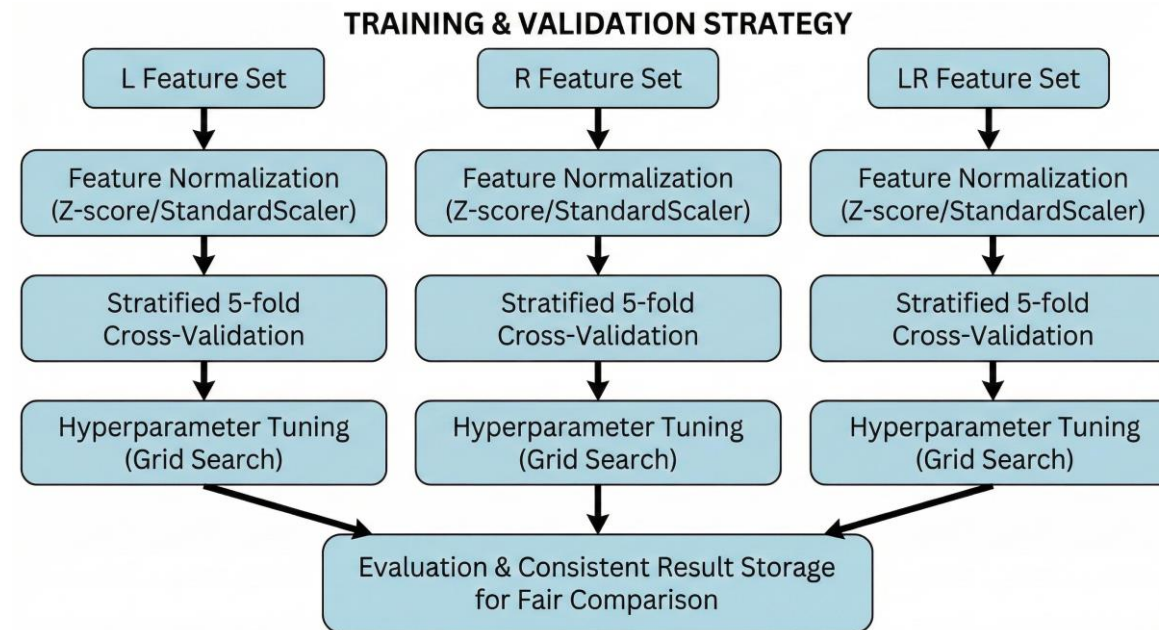


Figure: Training and validation pipeline using z-score normalization, stratified 5-fold cross-validation, and grid-search hyperparameter tuning for L, R, and LR feature sets.

Training Strategy:

- ✓ Z-score feature normalization using StandardScaler
- ✓ Stratified 5-fold cross-validation to preserve class balance
- ✓ Grid-search hyperparameter tuning on training folds
- ✓ Identical procedure applied to Left (L), Right (R), and Combined (LR) inputs
- ✓ Unified evaluation pipeline to ensure fair model comparison

Performance Metrics Used

Comprehensive Baseline Training & Evaluation Pipeline

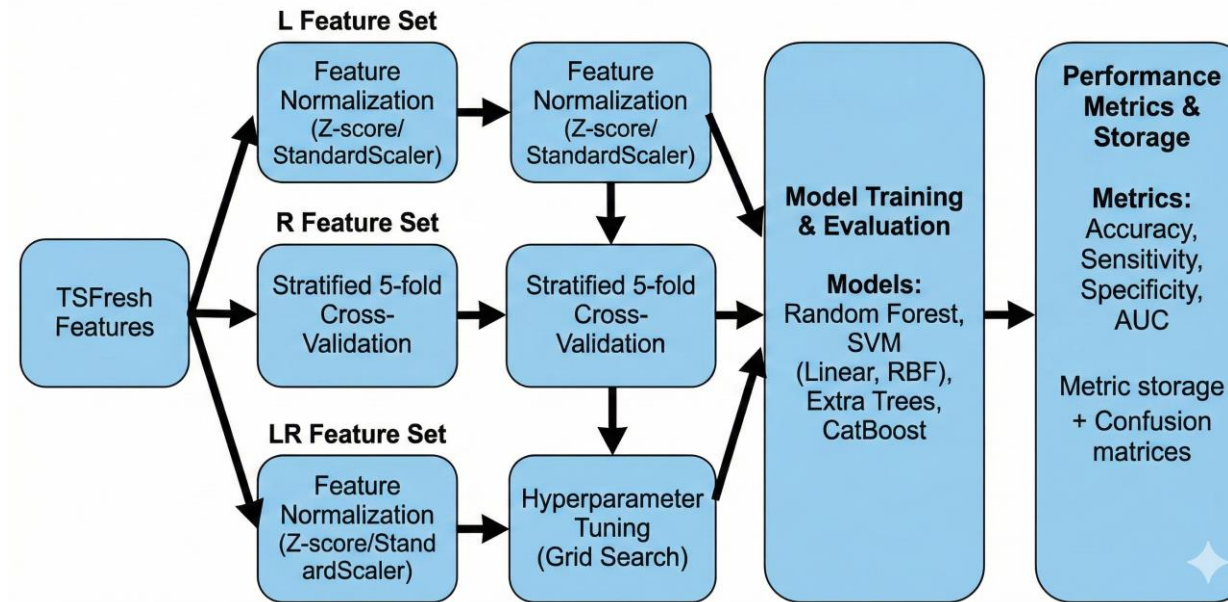


Figure: Performance metrics computation and evaluation workflow for baseline models.

Evaluation Metrics:

- ✓ Accuracy: overall classification correctness.
- ✓ Sensitivity: ability to correctly detect Parkinson's patients.
- ✓ Specificity: ability to correctly identify healthy controls.
- ✓ AUC: model's ability to separate classes across thresholds.

Metric-wise Model Comparison

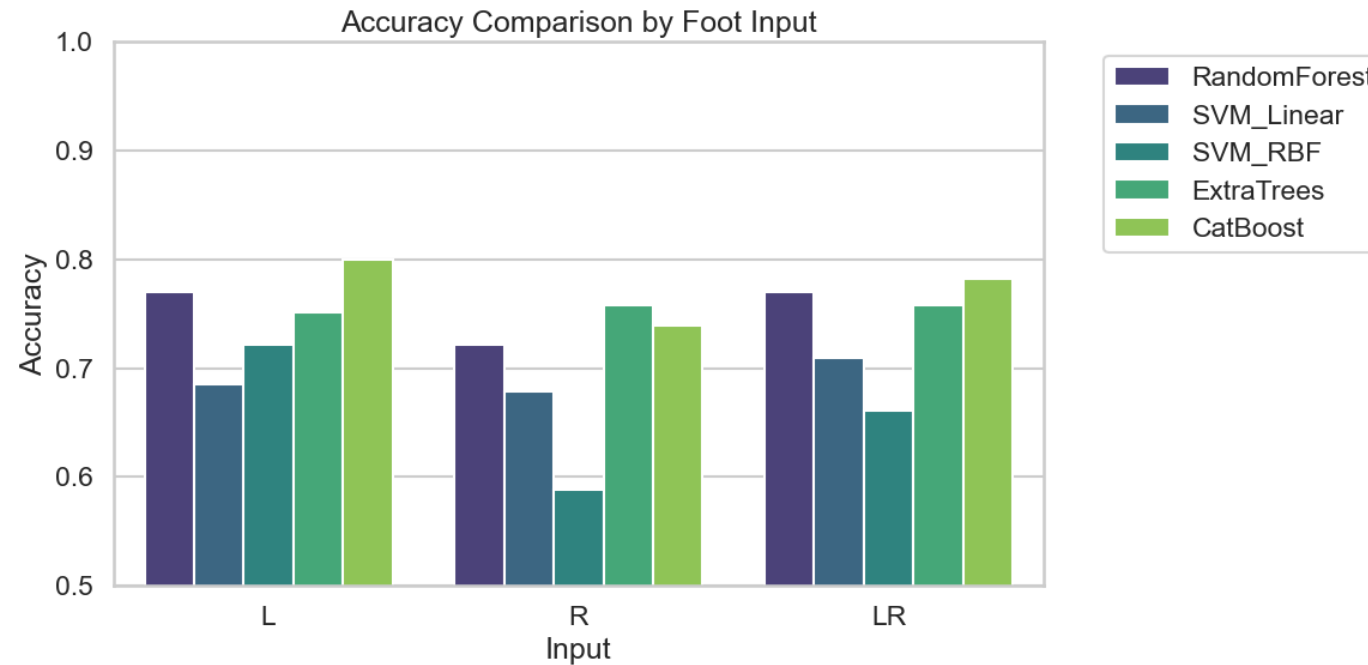


Figure: Accuracy comparison of baseline models across left (L), right (R), and combined (LR) foot input feature sets.

- ✓ Tree-based models outperform SVMs across all inputs.
- ✓ **CatBoost** achieves the highest accuracy for **L** and **LR**.
- ✓ Right-foot (**R**) input shows lower accuracy overall.
- ✓ Combining both feet (**LR**) improves performance stability.
- ✓ Accuracy varies with input modality and model type.

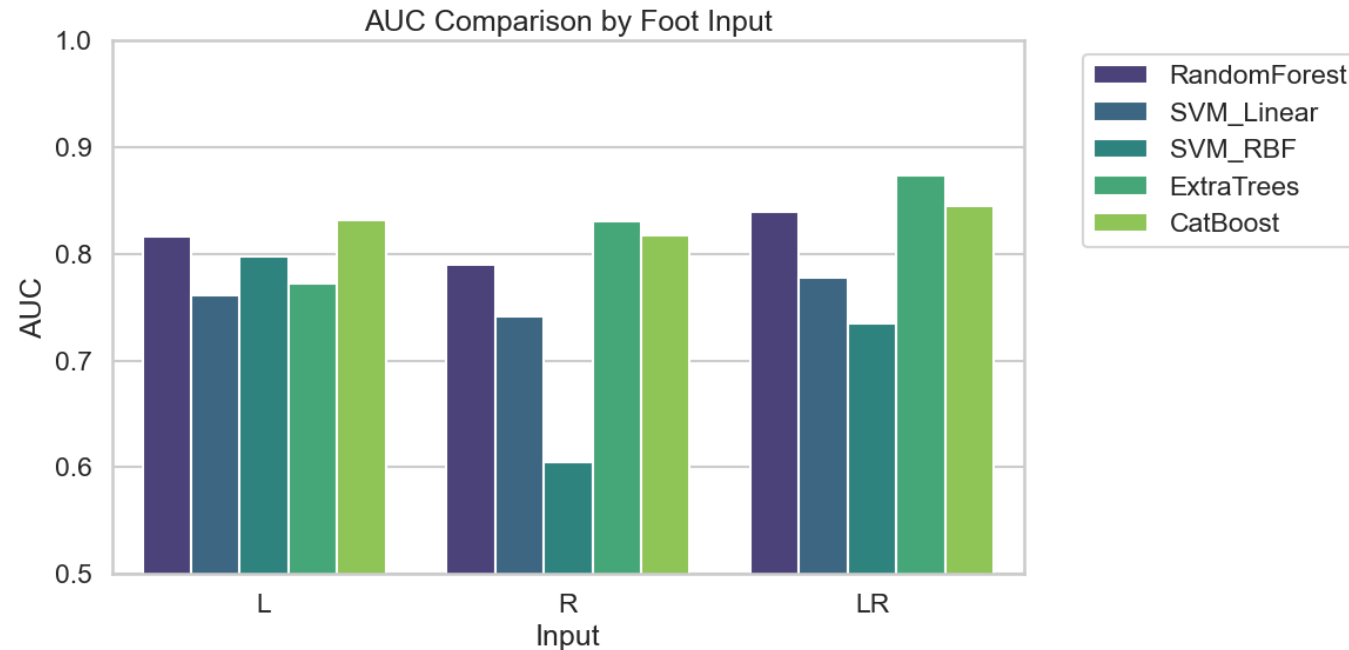


Figure: AUC comparison of baseline models across left (L), right (R), and combined (LR) foot input feature sets.

- ✓ **ExtraTrees** and **CatBoost** achieve the highest AUC across inputs.
- ✓ Combined input (**LR**) consistently improves class separability.
- ✓ SVM-RBF shows weaker AUC, especially for **R** input.
- ✓ Tree-based models provide more robust discrimination.
- ✓ AUC confirms ensemble methods as stronger baselines.

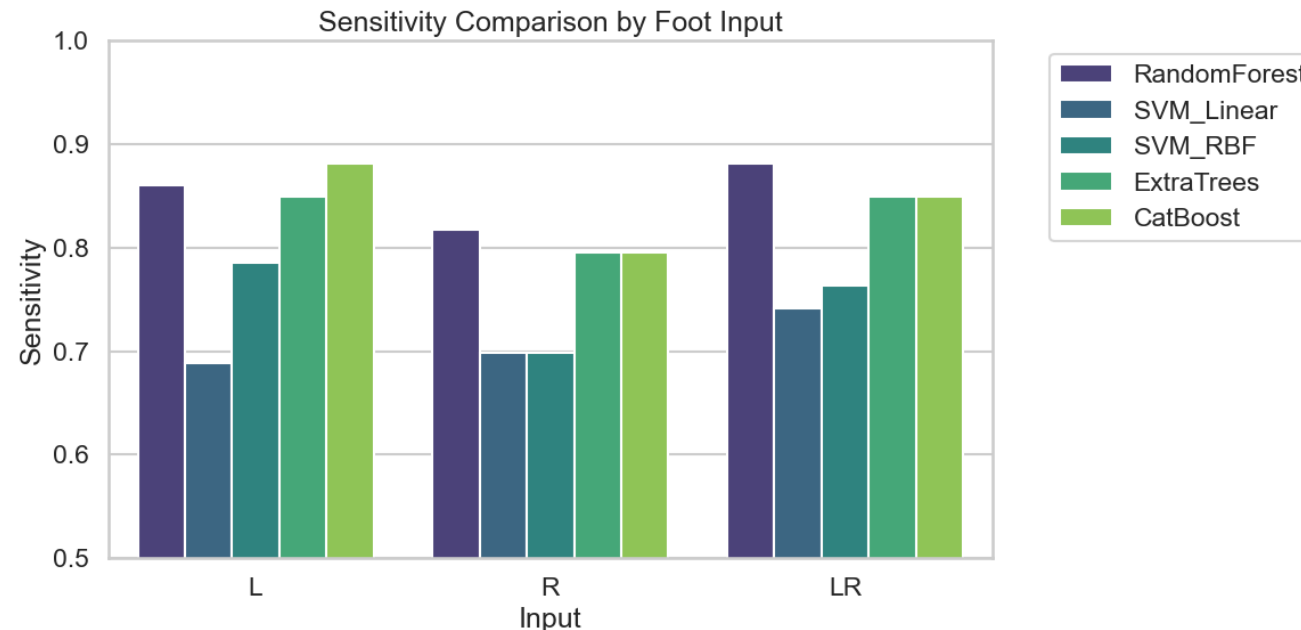


Figure: Sensitivity comparison of baseline models across left (L), right (R), and combined (LR) foot input feature sets.

- ✓ Tree-based models show **higher sensitivity** across all inputs.
- ✓ **RandomForest** and **CatBoost** detect PD cases most reliably.
- ✓ **LR (combined)** input improves sensitivity for all models.
- ✓ SVM-based models show comparatively lower recall.
- ✓ High sensitivity is critical for PD screening tasks.

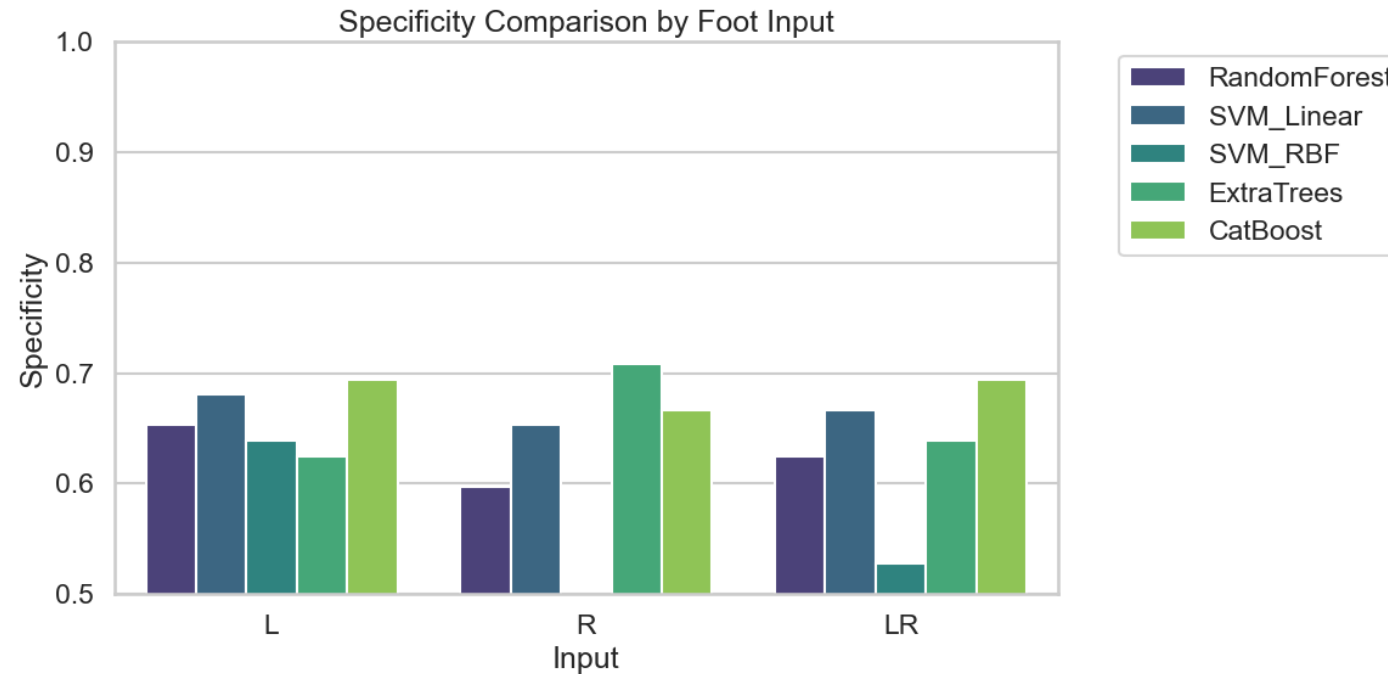


Figure: Specificity comparison of baseline models across left (L), right (R), and combined (LR) foot input feature sets.

- ✓ **SVM-Linear** and **CatBoost** achieve higher specificity.
- ✓ Tree models trade some specificity for sensitivity.
- ✓ **Right foot (R)** improves control-class detection.
- ✓ **LR input** provides a balanced specificity trend.
- ✓ Highlights the sensitivity–specificity trade-off.

Heatmap Overviews

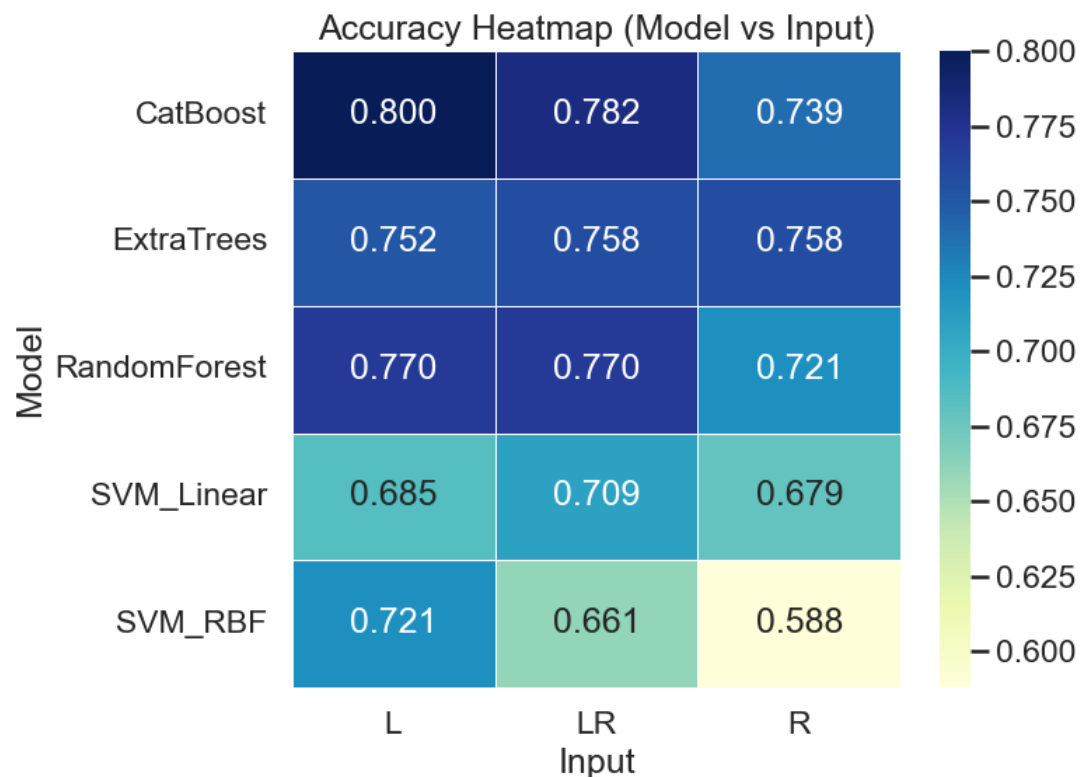


Figure: Accuracy heatmap comparing baseline models across left (L), right (R), and combined (LR) foot input feature sets.

- ✓ Tree-based models (**CatBoost**, **ExtraTrees**, **Random Forest**) consistently achieve higher accuracy across all inputs.
- ✓ **Left (L)** and **combined (LR)** inputs outperform right-only data for most models.
- ✓ **SVM-RBF** shows the weakest accuracy, especially on the **right foot (R)** input.

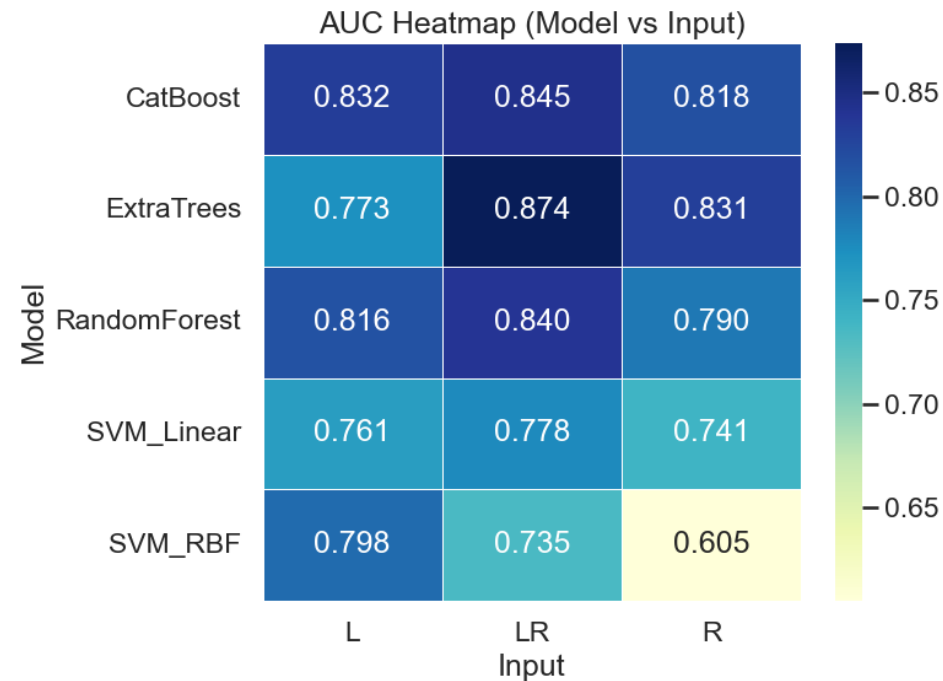


Figure: AUC heatmap comparing baseline models across left (L), right (R), and combined (LR) foot input feature sets.

- ✓ **ExtraTrees (LR)** achieves the highest AUC (0.87), followed closely by **CatBoost** and **Random Forest**, indicating strong discriminative performance.
- ✓ **Combined (LR)** and **left (L)** inputs consistently outperform right-only data across all models.
- ✓ **SVM-RBF** shows a clear drop in AUC on the right foot, highlighting its sensitivity to input choice.

Radar Plots (Model Behaviour Shape)

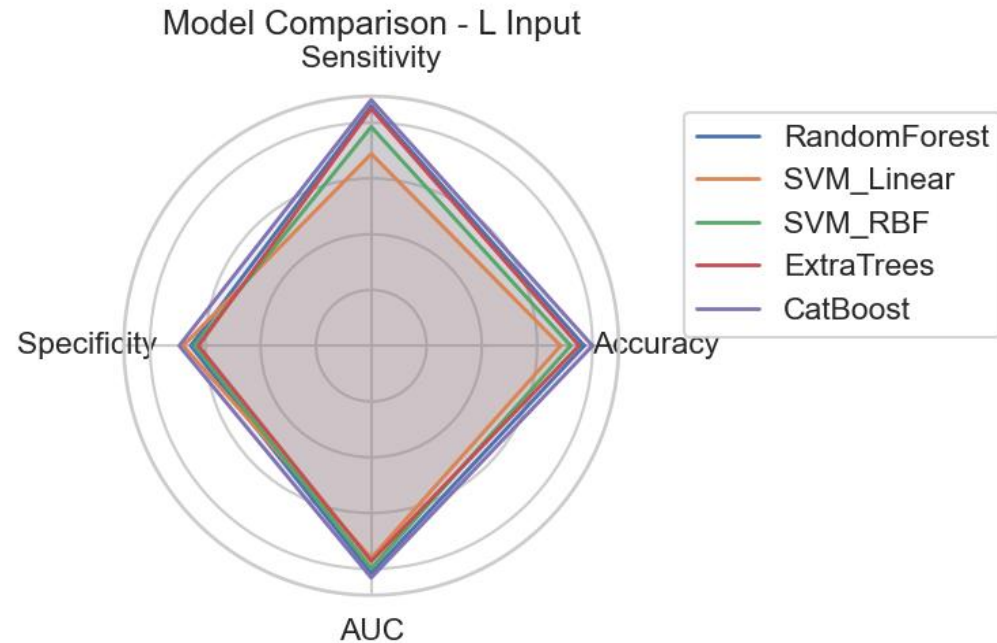


Figure: Radar plot illustrating the balanced performance profile of baseline models across accuracy, sensitivity, specificity, and AUC for the left-foot (L) input.

- ✓ For **left-foot input**, **CatBoost** and **ExtraTrees** show the most balanced performance across all metrics.
- ✓ **Random Forest** achieves strong sensitivity and AUC but slightly lower specificity.
- ✓ **SVM-based models** exhibit more uneven shapes, indicating less stable metric balance.

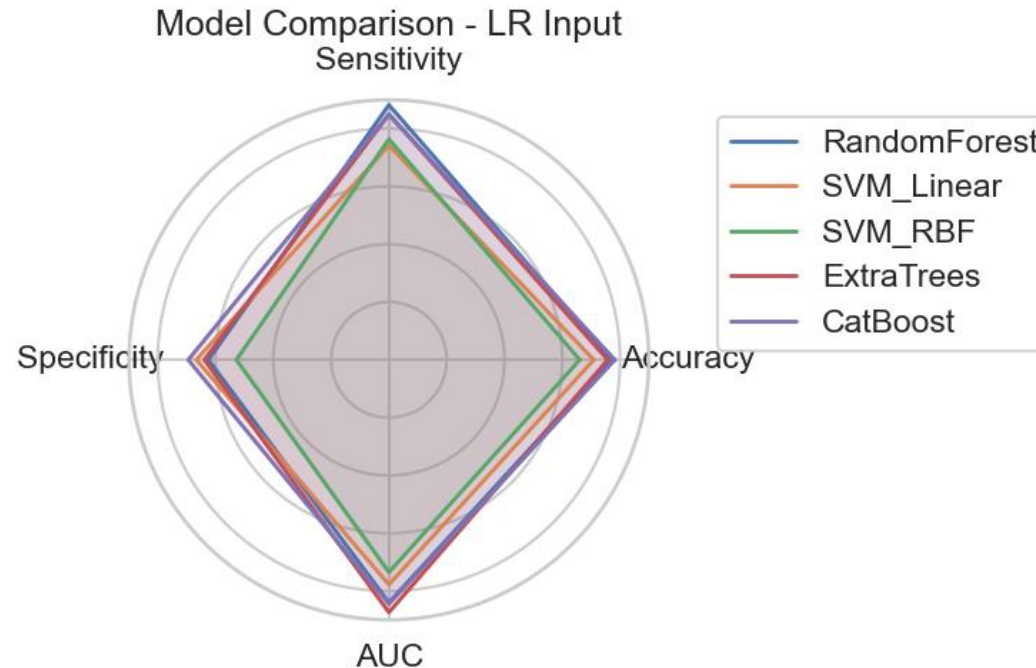


Figure: Radar plot comparing model behavior and metric balance across accuracy, sensitivity, specificity, and AUC for the combined left-right (LR) input features.

- ✓ For **combined (LR) input**, **CatBoost** and **ExtraTrees** again show the most uniform and balanced metric profiles.
- ✓ **Random Forest** achieves high sensitivity and AUC, with moderate specificity.
- ✓ **SVM models** remain less balanced, particularly in specificity, compared to tree-based methods.

Sensitivity vs Specificity Trade-Off

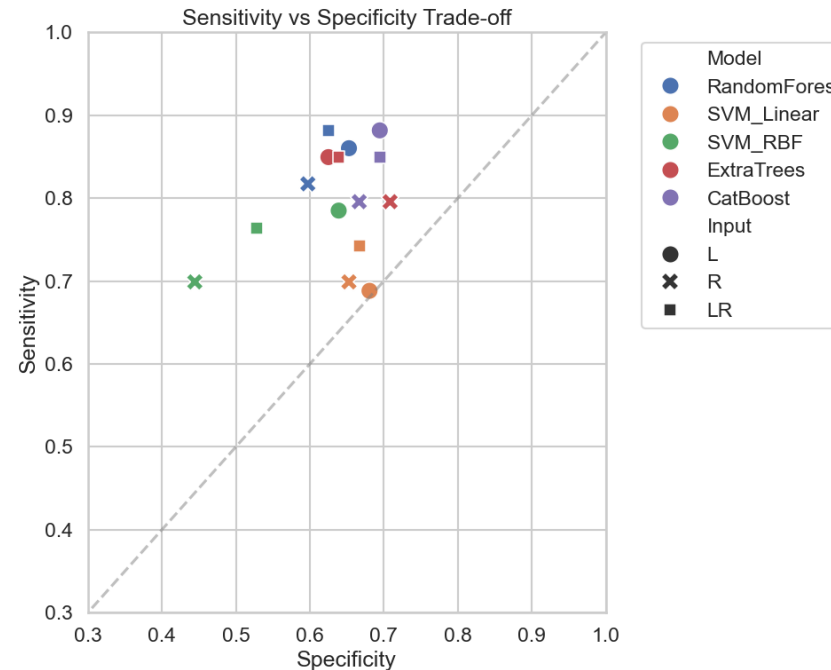


Figure: Sensitivity–specificity trade-off illustrating the balance between true positive and true negative rates across models and input feature configurations.

- ✓ The plot highlights the **trade-off between sensitivity and specificity** across all models and input types.
- ✓ **Tree-based models (CatBoost, ExtraTrees, Random Forest)** cluster closer to the upper-right region, indicating a better balance.
- ✓ **SVM models**, especially SVM-RBF, tend to sacrifice specificity to achieve higher sensitivity.

Performance Trend Across Input Types

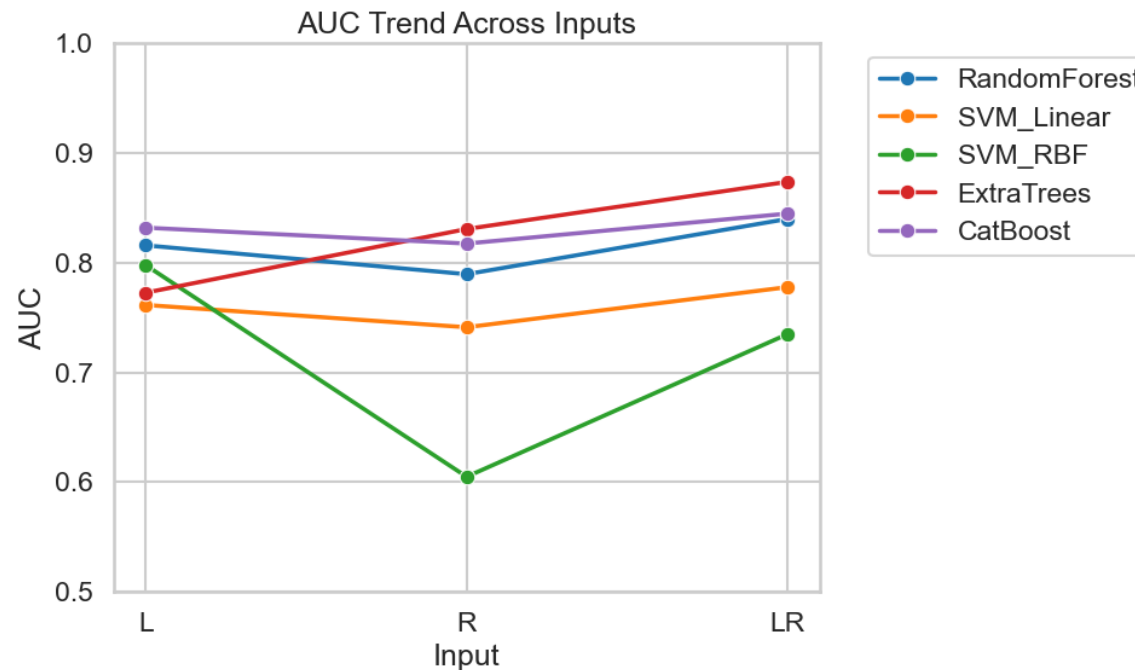


Figure: AUC trends across left (L), right (R), and combined (LR) foot inputs, showing improved discriminative performance when bilateral features are used.

- ✓ AUC generally **improves when combining left and right inputs (LR)** across most models.
- ✓ **Tree-based models (ExtraTrees, CatBoost, Random Forest)** show stable and consistently high AUC trends.
- ✓ **SVM-RBF is sensitive to input type**, dropping on right-only data but recovering with combined features.

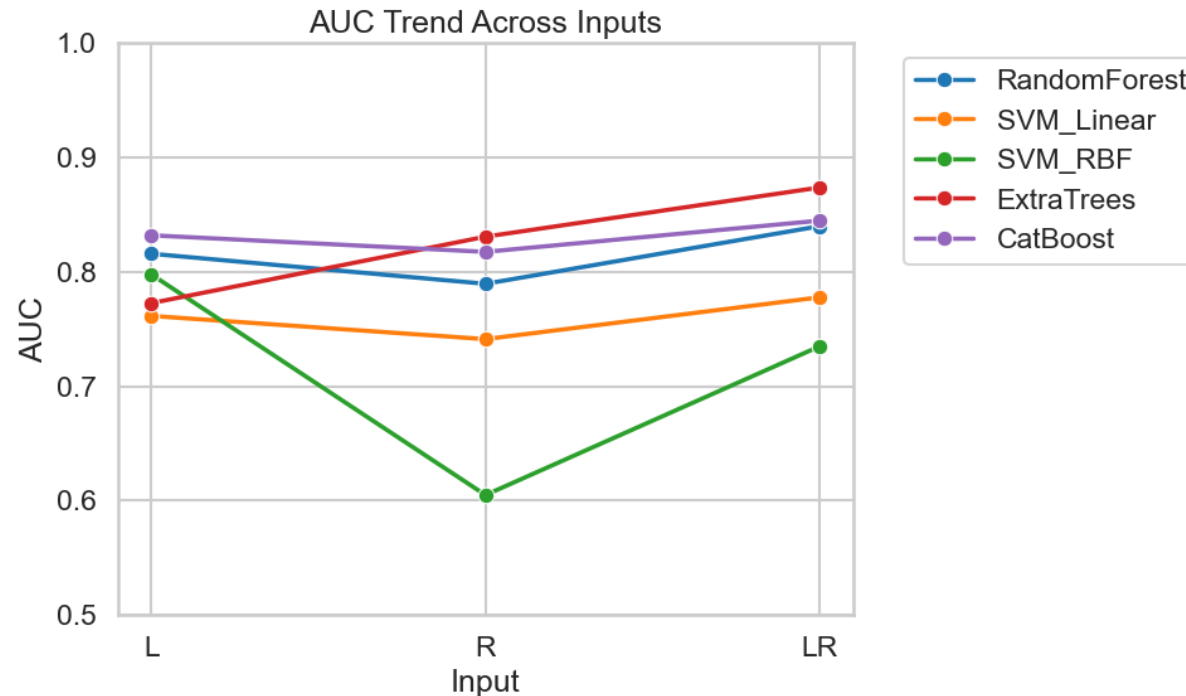


Figure: AUC trends across left (L), right (R), and combined (LR) inputs, showing improved performance with feature fusion and greater robustness of tree-based models.

- ✓ AUC values are **consistently higher for LR (combined) inputs** compared to single-foot inputs.
- ✓ **ExtraTrees and CatBoost show the strongest upward trend**, indicating robustness to feature fusion.
- ✓ **SVM-RBF is highly input-dependent**, with a clear drop for R-only input and recovery for LR.

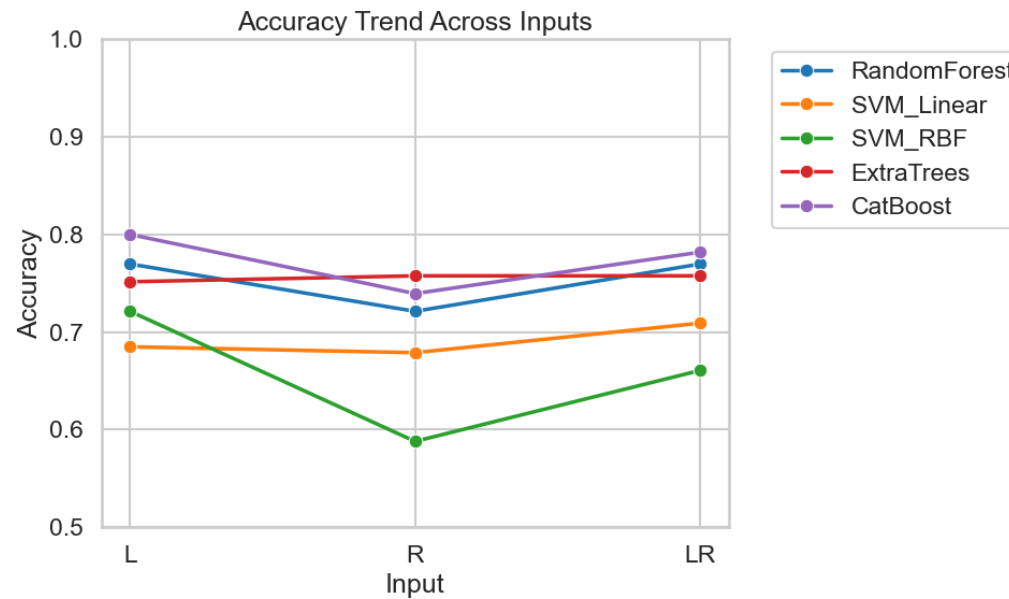


Figure: Accuracy trends across left (L), right (R), and combined (LR) inputs, showing improved stability and performance with feature fusion.

- ✓ Accuracy generally **improves when combining left and right foot inputs (LR)** across all models.
- ✓ **CatBoost and ExtraTrees achieve the most stable accuracy**, showing minimal performance drop across inputs.
- ✓ **SVM-RBF is the most sensitive to input choice**, with a clear degradation for R-only input.

Confusion Matrices

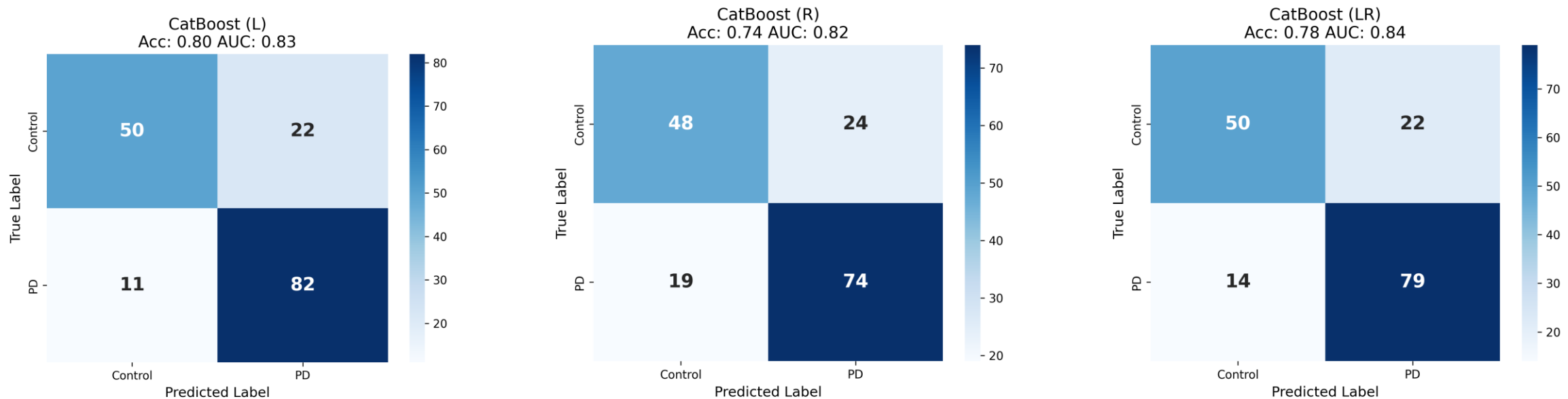


Figure: Confusion matrices for CatBoost across left (L), right (R), and combined (LR) inputs, illustrating improved PD detection and reduced false negatives with feature fusion.

- ✓ CatBoost shows **strong and balanced performance** across all input types.
- ✓ **LR input reduces false negatives** compared to single-foot inputs, improving PD detection reliability.
- ✓ High true-positive counts confirm CatBoost's **robust sensitivity without major loss in specificity**.

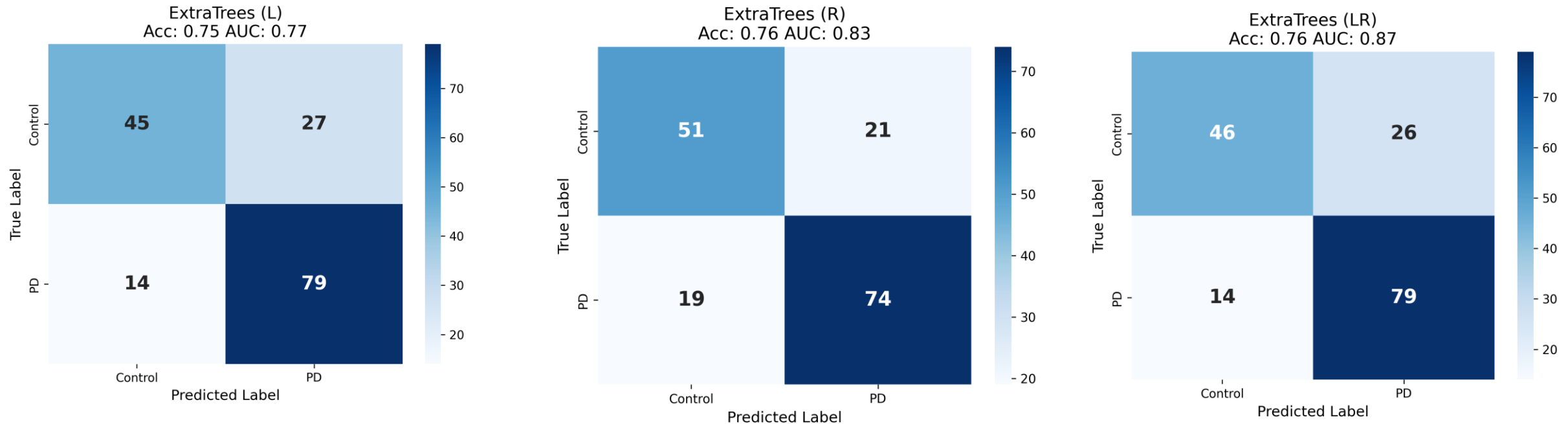


Figure: Confusion matrices for ExtraTrees across left (L), right (R), and combined (LR) inputs, demonstrating consistent PD detection performance and improved class balance with feature fusion.

- ✓ ExtraTrees delivers **consistently strong PD detection** across L, R, and LR inputs.
- ✓ **LR input achieves the best trade-off**, showing high true positives with stable false positives.
- ✓ Results highlight ExtraTrees' **robust generalization and low variance** compared to simpler models.

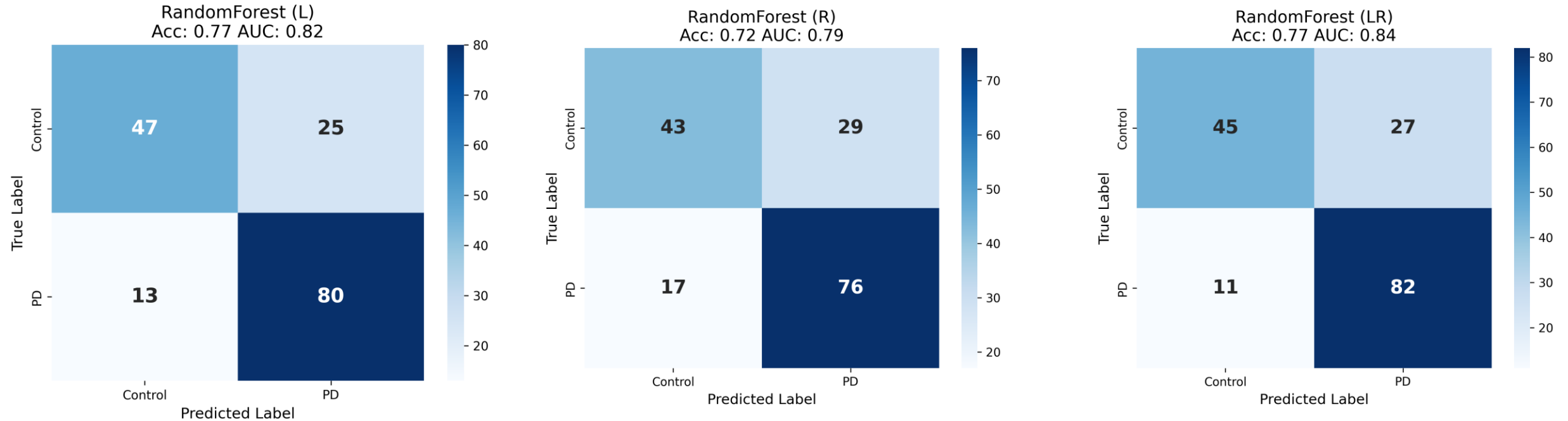


Figure: Confusion matrices for Random Forest across left (L), right (R), and combined (LR) inputs, illustrating stable performance and improved PD detection when left and right foot features are fused.

- ✓ Random Forest shows **stable and balanced performance** across L, R, and LR inputs.
- ✓ **LR input performs best**, achieving the highest AUC and strongest PD true-positive rate.
- ✓ Right-foot input slightly reduces accuracy, indicating **complementary benefit from foot fusion**.

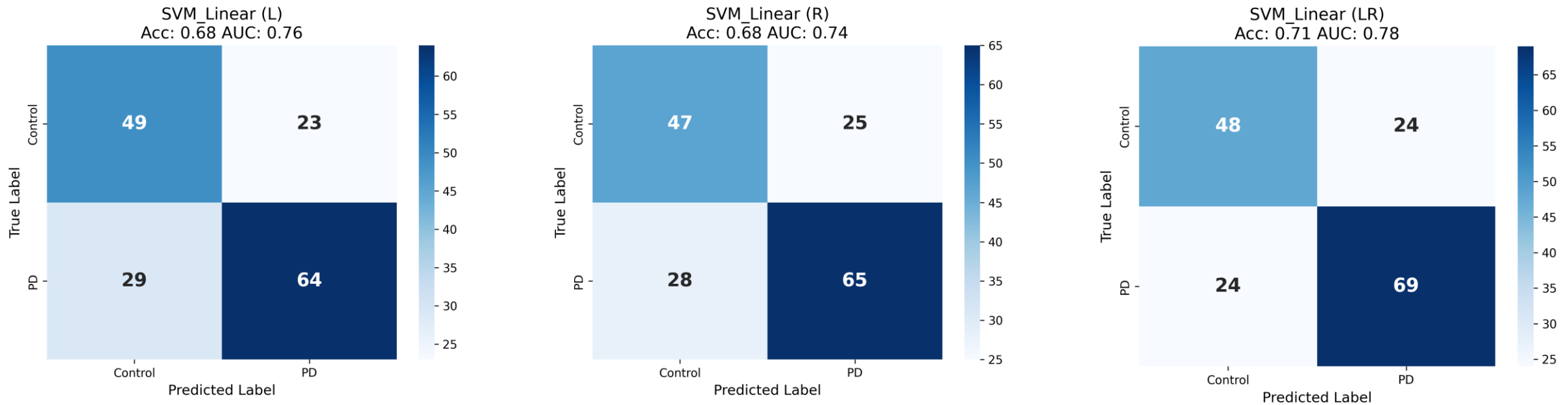


Figure: Confusion matrices for Linear SVM across left (L), right (R), and combined (LR) inputs, illustrating moderate classification performance and improved PD detection when left and right foot features are combined.

- ✓ Linear SVM shows **moderate performance**, with lower PD detection compared to tree-based models.
- ✓ **LR input improves results**, increasing both accuracy and AUC over single-foot inputs.
- ✓ Higher false negatives indicate **limited capacity to model complex gait patterns** with a linear boundary.

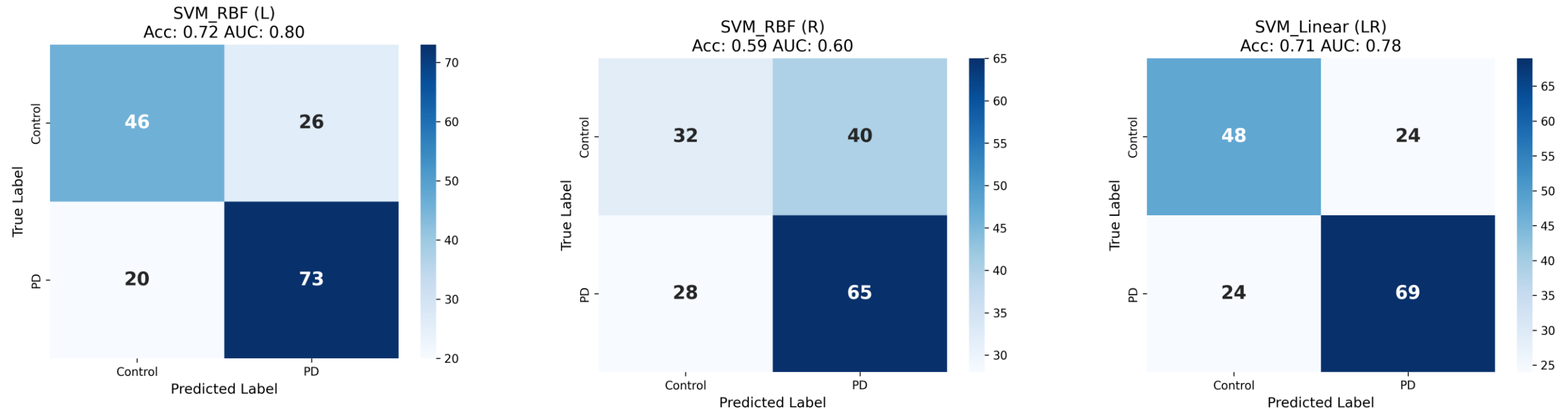


Figure: Confusion matrices for SVM-RBF across left (L), right (R), and combined (LR) inputs, highlighting strong performance on left-foot features, a significant drop for right-foot input, and partial recovery with feature fusion.

- ✓ SVM-RBF shows **strong performance for left-foot input**, indicating non-linear gait patterns are informative.
- ✓ Performance **drops significantly for right-foot input**, suggesting sensitivity to input variability.
- ✓ Combining both feet (LR) **stabilizes accuracy and AUC**, but still underperforms compared to tree-based models.

Interpretation & Key Insights

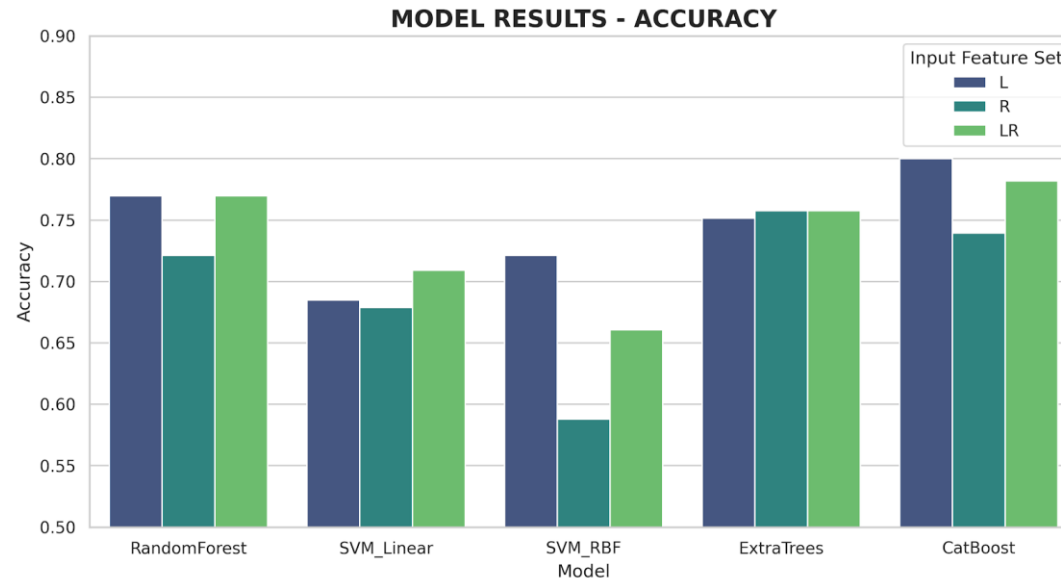


Figure: Summary of baseline model performance across input types, highlighting the impact of left–right (LR) feature fusion and differences in robustness between tree-based ensembles and SVM models.

- ✓ CatBoost achieves the highest overall accuracy, demonstrating strong and consistent classification performance across input types.
- ✓ ExtraTrees and Random Forest show stable accuracy trends, particularly when left and right foot features are combined (LR).
- ✓ Left–Right (LR) feature fusion generally improves accuracy robustness compared to single-foot inputs.
- ✓ SVM-based models exhibit lower accuracy than tree-based ensembles, indicating limited capacity to model complex gait patterns.
- ✓ Overall, tree-based models outperform SVMs on TSFresh features by better capturing non-linear feature interactions.

Model	Input	Accuracy	Sensitivity	Specificity	AUC
CatBoost	L	0.8	0.882	0.694	0.832
CatBoost	R	0.739	0.796	0.667	0.818
CatBoost	LR	0.782	0.849	0.694	0.845
ExtraTrees	L	0.752	0.849	0.625	0.773
ExtraTrees	R	0.758	0.796	0.708	0.831
ExtraTrees	LR	0.758	0.849	0.639	0.874
RandomForest	L	0.77	0.86	0.653	0.816
RandomForest	R	0.721	0.817	0.597	0.79
RandomForest	LR	0.77	0.882	0.625	0.84
SVM_Linear	L	0.685	0.688	0.681	0.761
SVM_Linear	R	0.679	0.699	0.653	0.741
SVM_Linear	LR	0.709	0.742	0.667	0.778
SVM_RBF	L	0.721	0.785	0.639	0.798
SVM_RBF	R	0.588	0.699	0.444	0.605
SVM_RBF	LR	0.661	0.763	0.528	0.735

Figure: Baseline performance comparison of classical machine learning models across Left (L), Right (R), and Combined (LR) gait inputs, evaluated using Accuracy, Sensitivity, Specificity, and AUC.

- ✓ Tree-based models (CatBoost, ExtraTrees, Random Forest) consistently outperform SVMs, showing stronger capability in capturing complex gait patterns.
- ✓ Combining left and right foot signals (LR) improves overall robustness and AUC, indicating complementary gait information from both feet.
- ✓ CatBoost achieves the highest single-input accuracy (~0.80) using left-foot features, demonstrating strong classification capability.
- ✓ ExtraTrees with combined inputs achieves the best overall separability (AUC $\approx$ 0.87), highlighting the benefit of feature fusion.

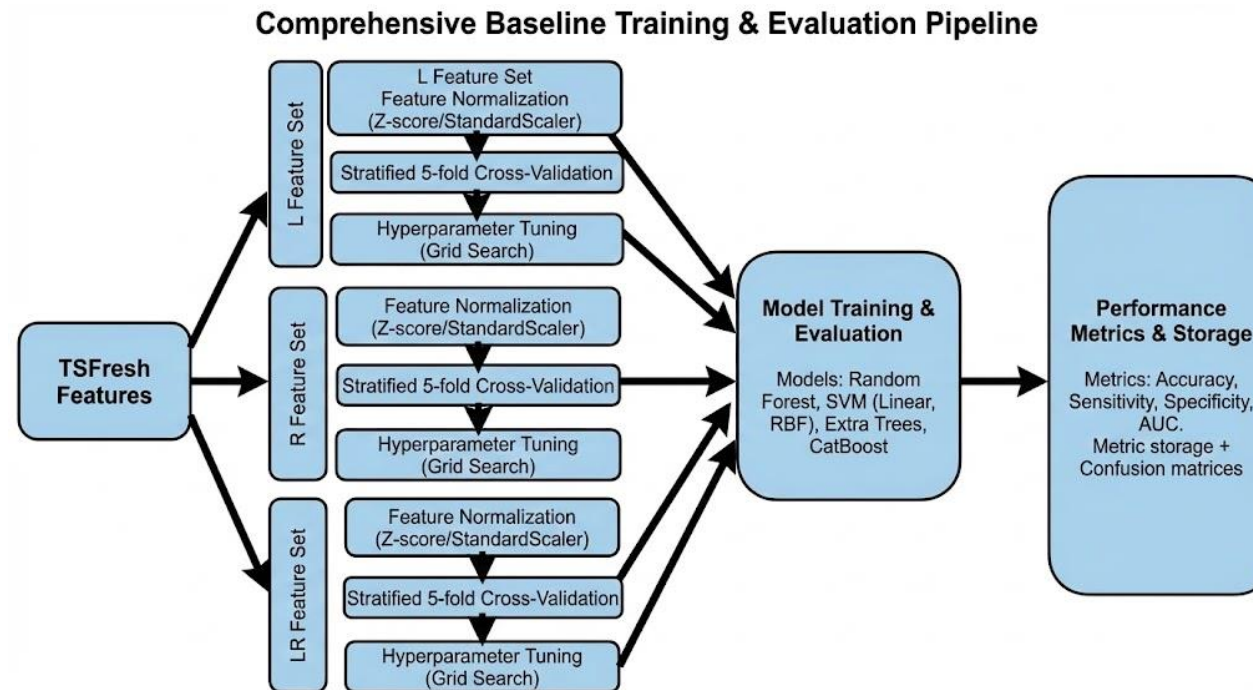


Figure: Baseline training and evaluation pipeline using TSFresh features with stratified cross-validation, hyperparameter tuning, and multi-metric evaluation across L, R, and LR inputs.

- ✓ **Robust machine learning baselines established** using TSFresh gait features.
- ✓ **Five classifiers evaluated** under stratified 5-fold cross-validation with hyperparameter tuning.
- ✓ **Comprehensive evaluation conducted**, including 15 confusion matrices and 11 comparative analyses.
- ✓ **Tree-based ensemble models (ExtraTrees, CatBoost)** deliver the strongest and most stable performance.
- ✓ The results **provide a reliable benchmark for gait-based Parkinson's disease classification**.

Discussion & Next Steps

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- ✓ Tree-based models show strong baseline performance, confirming that TSFresh features contain discriminative gait information for PD detection.
- ✓ Feature-based models ignore temporal dynamics, suggesting potential benefits from sequence-aware models.
- ✓ Future work: evaluate deep learning architectures (RNN, Transformer, or State-Space models) that directly learn temporal patterns from gait signals.
- ✓ Model interpretability: analyze feature importance and gait biomarkers to better understand clinical relevance.
- ✓ Larger datasets and cross-dataset validation are needed to ensure robustness and clinical applicability.

Q & A

Thank you

